

this variation is that climographs are based on annual *averages*, but the *pattern* of climatic variation is often as important as the average climate. For example, some areas may receive regular precipitation throughout the year, whereas other areas may have distinct wet and dry seasons.

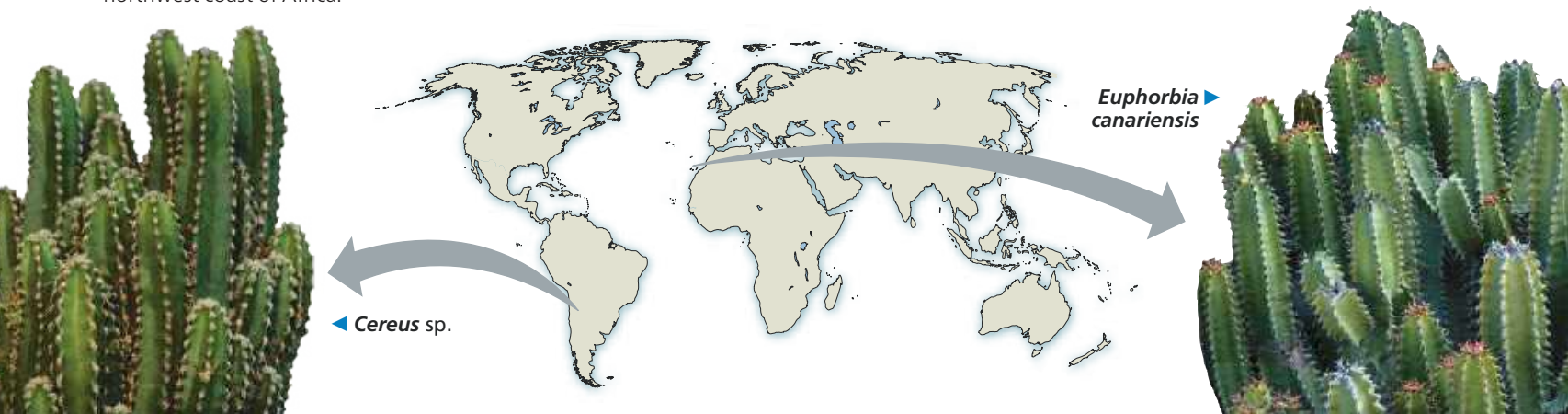
General Features of Terrestrial Biomes

Most terrestrial biomes are named for major climatic features and for their predominant vegetation. Temperate grasslands, for instance, are generally found in middle latitudes, where the climate is moderate, and are dominated by various grass species. Each biome is also characterized by microorganisms, fungi, and animals adapted to that particular environment. Temperate grasslands are usually more likely than temperate broadleaf forests to be populated by arbuscular mycorrhizal fungi (see Figure 37.14) and by large grazing mammals. Although Figure 52.10 shows distinct boundaries between the biomes, terrestrial biomes usually grade into neighboring biomes, sometimes over large areas. The area of intergradation, called an **ecotone**, may be wide or narrow.

Vertical layering of vegetation is an important feature of terrestrial biomes. In many forests, the layers from top to bottom consist of the upper **canopy**, the low-tree layer, the shrub understory, the ground layer of herbaceous plants, the forest floor (litter layer), and the root layer. Nonforest biomes have similar, though usually less pronounced, layers. Layering of vegetation provides many different habitats for animals, which sometimes exist in well-defined feeding groups, from the insectivorous birds and bats that feed above canopies to the worms, arthropods, and small mammals that search for food in the litter and root layers below.

The species composition of each kind of biome varies from place to place. For instance, in the northern coniferous forest (taiga) of North America, red spruce is common in the east but does not occur in most other areas, where black spruce and white spruce are abundant. As **Figure 52.12** shows, cacti living in deserts of North and South America resemble plants called euphorbs found in African deserts. But since cacti and euphorbs belong to different evolutionary lineages, their similarities are due to convergent evolution rather than shared ancestry.

▼ **Figure 52.12 Convergent evolution in a cactus and a euphorb.** Cacti in the genus *Cereus* are found in the Americas; *Euphorbia canariensis*, a euphorb, is native to the Canary Islands, off the northwest coast of Africa.



Disturbance and Terrestrial Biomes

Biomes are dynamic, and disturbance rather than stability tends to be the rule. In ecological terms, **disturbance** is an event such as a storm, fire, or human activity that changes a community, removing organisms from it and altering resource availability. For example, hurricanes and other storms create openings for new species in many tropical and temperate forests and can alter forest composition. After Hurricane Katrina struck the Gulf coast of the United States in 2005, mixed swamp forests in the area shifted toward a dominance of bald cypress (*Taxodium distichum*) and water tupelo (*Nyssa aquatica*) because these species are less susceptible to wind damage than other tree species found there. As a result of disturbances, biomes are often patchy, containing several different communities in a single area.

In many biomes, even the dominant plants depend on periodic disturbance. Natural wildfires are an integral component of grasslands, savannas, chaparral, and many coniferous forests. Before agricultural and urban development, much of the southeastern United States was dominated by a single conifer species, the longleaf pine. Without periodic burning, broadleaf trees tended to replace the pines. Forest managers now use fire as a tool to help maintain many coniferous forests.

Figure 52.13 summarizes the major features of terrestrial biomes. As you read about the characteristics of each biome, remember that humans have altered much of Earth's surface, replacing natural communities with urban and agricultural ones. The central United States, for example, is classified as grassland and once contained extensive areas of tallgrass prairie. Very little of the original prairie remains today, however, having been converted to farmland.

CONCEPT CHECK 52.2

1. Based on the climograph in Figure 52.11, what mainly differentiates temperate grassland from temperate broadleaf forest?
2. Using Figure 52.13, identify the natural biome in which you live, and summarize its abiotic and biotic characteristics. Do these reflect your actual surroundings? Explain.
3. **WHAT IF?** If average temperatures on Earth increase by 4°C in this century, predict which biome would be most likely to replace tundra in some locations as a result. Explain.

For suggested answers, see Appendix A.